

L'Hôpital's rule — 0.4.8

$$\lim_{x \rightarrow \infty} \frac{x^2}{x} = \infty$$

$$\lim_{x \rightarrow \infty} \frac{2x}{1} = \infty$$

$$\lim_{x \rightarrow \infty} \frac{x}{x^2} = 0$$

$$\lim_{x \rightarrow \infty} \frac{1}{2x} = 0$$

Theorem (L'Hôpital's rule): Suppose that $\lim_{x \rightarrow a} f(x) = 0$

and $\lim_{x \rightarrow a} g(x) = 0$ and that $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists.

Then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ also exists and equals $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.

(And the same is true if $f(x), g(x) \rightarrow \pm\infty$ as $x \rightarrow a$).

If we additionally assume that f' and g' are continuous at a and $g'(a) \neq 0$, then we can prove the theorem as follows:

$$\begin{aligned}
 \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} \cdot \frac{(x-a)}{(x-a)} \\
 &= \lim_{x \rightarrow a} \frac{\frac{f(x) - f(a)}{x-a}}{\frac{g(x) - g(a)}{x-a}} = \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x-a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x-a}} = \frac{f'(a)}{g'(a)} = \frac{\lim_{x \rightarrow a} f'(x)}{\lim_{x \rightarrow a} g'(x)} \\
 &= \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.
 \end{aligned}$$

Example: $\lim_{x \rightarrow 1} \frac{\sin(\pi x)}{\ln(x)} \xrightarrow{0/0} \overset{L'H}{=} \lim_{x \rightarrow 1} \frac{\pi \cos(\pi x)}{\frac{1}{x}}$

$$= \lim_{x \rightarrow 1} x \pi \cos(\pi x) = \pi \cos(\pi) = -\pi$$